

The core idea to carry into the visit

Fjernvarme Fyn is **not merely buying an AI model or some cameras**. They are buying an operational system that must:

See the incoming load, associate it with the correct truck, detect problematic waste, notify the right operator in time, fit safely into the existing plant, and remain maintainable without dependence on ARIS.

The formal purpose is to monitor, identify and classify incoming waste so they can reduce the receipt and handling of hazardous or unauthorized waste. The evaluation also considers much more than detection accuracy: price, requirement compliance, usability, implementation, transition to operations and support. The documents do not publish scoring weights, so you should not claim that Appendix 1 is definitively the largest scoring element.

Memorize it as LOADS

L — Load and operational workflow

Understand the journey of one load:

1. The truck arrives.
2. The truck and license plate are identified.
3. The truck reaches an unloading position.
4. Waste becomes visible before, during or after unloading.
5. The system detects objects or evaluates the overall load.
6. An alarm reaches an operator.
7. The operator decides whether to inspect, reject, register or otherwise handle the load.
8. The event and associated evidence are stored and searchable.

Your most important operational questions are therefore:

- At what point should detection occur?
- Is the decision made per object, per load, or both?
- How is a detection connected to the correct truck and weighing transaction?
- Who receives the alarm?
- What can that person realistically do after receiving it?
- How quickly must the alarm arrive to still be useful?

Do not treat “real-time” as a technical frame rate. Operationally, it means **early enough for someone to take an effective action**.

The required waste types are easier to remember in three groups:

Waste-quality problems

- Plasterboard
- Mineral wool
- Generally low-calorific-value loads

Potentially hazardous or unauthorized material

- PVC-containing plastics
- Electronic waste
- Gas cylinders
- Batteries

Operationally troublesome material

- Copper and metal fractions
- Large or bulky items
- Rolling objects such as whole tyres or plastic rolls
- Opaque bags

POPs are identified as a possible future capability rather than a present detection requirement.

A particularly good question is:

“Could you walk us through one complete load—from arrival and weighing through unloading and operator response—and show us exactly where a detection would still be actionable?”

That sounds customer-focused because it starts with their process rather than ARIS’s technology.

O — Optics and physical environment

Tomorrow, your job is to determine whether the requested detections are physically observable.

Look for:

- Truck routes and stopping positions
- Number of possible simultaneous unloading positions
- Whether the licence plate is visible when needed
- When the waste is most spread out and least occluded
- Whether the system can see into the truck, falling waste or only the final pile
- Crane activity and other recurring obstructions
- Structures suitable for mounting cameras
- Mounting height, distance and angle
- Blind spots and whether another camera could remove them
- Lighting variation, glare and deep shadows
- Dust, smoke, steam and airborne debris
- Heat, vibration and impact exposure
- How quickly a camera window would become dirty
- Availability of compressed air for cleaning
- Power and network distances
- A safe location for edge computers, switches and enclosures
- Access requirements for maintenance

“Full coverage” needs clarification. It might mean:

- Every unloading position is observable.
- Every part of the waste surface is visible simultaneously.
- Every load is observed at some point.
- No blind spots are accepted.
- Blind spots are accepted if another camera sees the same material later.

Those interpretations result in very different camera counts and costs.

Do **not** promise a final camera count tomorrow. A stronger answer is:

“Based on the visit, we will define candidate coverage zones and identify which uncertainties require a survey, test recording or design-phase validation.”

A — Architecture, ABB and alarms

This is the most critical unresolved subject in the material.

Appendix 2 says:

- Signals, alarms and events must be transferable to DCS/SCADA.
- Operators must be able to monitor and operate relevant plant functions through ABB 800xA.
- Configurations and documentation must be open and maintainable.
- There must be no black-box restriction preventing configuration changes, integrations, repairs or a future change of supplier.

But Appendix 1 requirement 2.8 says the solution must be **fully physically and logically isolated**, with no direct or indirect connection to Fjernvarme Fyn's LAN, WAN, Wi-Fi, VPN or other internal communication services.

At the same time, the draft contract says the system must be supplier-operated SaaS and must not depend on Fjernvarme Fyn's internal servers.

Appendix 2 additionally says remote or cloud connections require written approval.

These statements cannot all be implemented literally without an agreed integration boundary. Possible interpretations include:

- An isolated ARIS network with a controlled one-way interface
- An industrial DMZ
- A protocol gateway operated by Fjernvarme Fyn
- An alarm-only interface to ABB 800xA
- A separate cellular connection for SaaS, plus a restricted OT gateway
- A completely separate system with operators manually acknowledging alarms

You need to establish:

- Which ABB 800xA version they use
- Which protocols or gateways are approved
- Whether ARIS provides only alarms/data or also HMI components
- Who performs ABB engineering
- Whether they already have an ABB or OT integration partner
- Whether bidirectional control is actually expected
- Where the security boundary is
- Who owns firewall, VLAN and SIEM configuration
- Whether remote maintenance will ever be permitted
- Whether the dashboard remains separate while only alarm summaries enter ABB

This one decision affects architecture, cybersecurity, responsibilities, price, support and feasibility.

D — Data, cybersecurity and GDPR

Remember the distinction:

- **Fjernvarme Fyn is the data controller.**
- ARIS would be the data processor when processing personal data on its behalf.

Potential personal data includes licence plates, identifiable people in images and user-account information.

Important requirements include:

- SAML 2.0 SSO
- Secure authentication and token exchange
- EU-only personal-data processing and storage
- Configurable retention and automatic deletion
- Downloadable images/video with timestamps and metadata
- Security logs or periodic access reports
- Patch management and infrastructure monitoring
- Prior authorization of subprocessors
- Support for DPIAs
- Deletion or return of data after termination
- No third-country transfers without documented authorization
- Breach notification without undue delay and, where possible, within 24 hours.

A major point: the DPA's Appendices A, B and C still contain placeholders for exact data categories, retention periods, processing locations, technical security measures and audit procedures. Those details are therefore **not yet settled**.

Ask specifically:

“May operational images containing licence plates or people be used for model improvement, and if so, under what retention, access, masking and processing-location conditions?”

Also correct one point in your notes: ARIS does not necessarily need an ISO 27001 certificate, but it **must demonstrate an information-security management system equivalent to ISO 27001**. That is a minimum eligibility requirement, not merely a desirable characteristic of the product. Without certification, ARIS must document its controls across access, assets, software development, backup, encryption, incident response, GDPR and physical security.

S — Scope, support, schedule and responsibilities

The tender is judged on whether the entire delivery is credible:

- Hardware
- Software/SaaS
- Installation and commissioning
- Integration
- Project management
- Training
- Documentation
- Support
- Long-term operation
- Exit and data migration

The offer must include architecture diagrams, installation requirements, a transparent project model, clear responsibilities, support routes and a detailed implementation plan covering milestones, dependencies, risks and testing.

The draft contract already proposes a more specific installation split than your notes suggest:

- Fjernvarme Fyn establishes compressed air, power, network connectivity, scaffolding and access arrangements.
- ARIS advises on camera placement and mounting.
- Fjernvarme Fyn performs the actual mounting under ARIS supervision.
- ARIS performs final testing and verification.

Ask whether this division is intentional and fixed. It materially changes ARIS's installation pricing and liability.

Four details you should correct before tomorrow

1. 1 February 2027 is not the delivery deadline

The ITT explicitly calls it “**Commencement for work on site.**” The final completion/hand-over date is not yet completed in the draft contract; it currently says “XX 2027.”

A precise question is:

“Can you confirm that 1 February is the start of onsite implementation, and that the completion and acceptance date will be established in Appendix 7?”

2. The delay penalties are inconsistent

Appendix 6 says 2.5% of the total order amount for every commenced week of delay.

The higher-priority draft SaaS contract says EUR 2,500 for every commenced calendar day after the hand-over deadline.

The contract document has precedence over the appendices, but the completion date is still a placeholder. The commercial team should treat this as a clarification item, not assume the weekly figure applies. The contractual precedence is contract, ITT, Appendix 1, Appendix 2 and then the remaining appendices.

3. Acceptance needs a measurable definition

You need to know what proves the system is accepted:

- Camera coverage
- Detection categories
- Precision or false-alarm rate
- Recall or missed-detection rate
- Minimum detectable object size
- Maximum alarm latency
- Licence-plate readability
- Availability
- Integration test
- Retention and export functions
- Operator training
- Documentation delivery

Avoid accepting a vague requirement such as “the system works satisfactorily.” Acceptance should be linked to agreed test cases and operating conditions.

4. The tender may be awarded without negotiation

They reserve the right to award based on the initial tender. ARIS cannot rely on negotiations to repair unclear reservations or shortcomings later. Reservations also create a material rejection risk.

How to leave a strong impression

Use this pattern throughout the visit:

1. **Ask about their process.**
2. **Observe the physical constraint.**
3. **Explain why it matters technically.**
4. **Confirm your understanding.**
5. **Avoid promising until the uncertainty is resolved.**

For example:

“We can see that waste is substantially more visible while it is falling than once it has formed a compact pile. Is the required decision expected during unloading, or is post-unloading inspection sufficient? That distinction will determine both camera placement and achievable detection performance.”

That demonstrates practical machine-vision understanding without turning the discussion into a sales pitch.

Bring the prescribed PPE: safety shoes, helmet, safety glasses, long trousers and a high-visibility vest or jacket. Also bring personal identification for access-card issuance.